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MULTI-FUNCTIONAL PERITONEAL DIALYSIS MOBILE CART

Abstract

A mobile dialysis therapy cart includes a top shelf and a fluid bag management compartment located beneath the top shelf. The fluid bag management compartment includes at least one heating device for heating a fluid bag. The mobile dialysis therapy cart also includes at least one temperature sensor located to sense a temperature associated with dialysis fluid located within the fluid bag. The mobile dialysis therapy cart further includes a control unit communicatively coupled to the at least one heating device and the at least one temperature sensor. The control unit is programmed to use an output from the at least one temperature sensor to operate the at least one heating device to heat the dialysis fluid within the fluid bag to a desired temperature.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a divisional application of U.S. patent application Ser. No. 17/331,158, filed on May 26, 2021, entitled “MULTI-FUNCTIONAL PERITONEAL DIALYSIS MOBILE CART”, which claims the benefit of and priority to U.S. Provisional Patent Application No. 63/030,502, filed on May 27, 2020, entitled “MULTI-FUNCTIONAL PERITONEAL DIALYSIS MOBILE CART”, the contents thereof are incorporated herein by reference in their entirety.

BACKGROUND

[0002] The present disclosure relates generally to medical fluid treatments and in particular to dialysis fluid treatments. Due to various causes, a person's renal system can fail. Renal failure produces several physiological derangements. It is no longer possible to balance water and minerals or to excrete daily metabolic load. Toxic end products of metabolism, such as, urea, creatinine, uric acid and others, may accumulate in a patient's blood and tissue.

[0003] Reduced kidney function and, above all, kidney failure is treated with dialysis. Dialysis removes waste, toxins and excess water from the body that normal functioning kidneys would otherwise remove. Dialysis treatment for replacement of kidney functions is critical to many people because the treatment is lifesaving.

[0004] One type of kidney failure therapy is Hemodialysis (“HD”), which in general uses diffusion to remove waste products from a patient's blood. A diffusive gradient occurs across the semi-permeable dialyzer between the blood and an electrolyte solution called dialysate or dialysis fluid to cause diffusion.

[0005] Hemofiltration (“HF”) is an alternative renal replacement therapy that relies on a convective transport of toxins from the patient's blood. HF is accomplished by adding substitution or replacement fluid to the extracorporeal circuit during treatment. The substitution fluid and the fluid accumulated by the patient in between treatments is ultrafiltered over the course of the HF treatment, providing a convective transport mechanism that is particularly beneficial in removing middle and large molecules.

[0006] Hemodiafiltration (“HDF”) is a treatment modality that combines convective and diffusive clearances. HDF uses dialysis fluid flowing through a dialyzer, similar to standard hemodialysis, to provide diffusive clearance. In addition, substitution solution is provided directly to the extracorporeal circuit, providing convective clearance.

[0007] Most HD, HDF, and HF treatments occur in centers. A trend towards home hemodialysis (“HHD”) exists today in part because HHD can be performed daily, offering therapeutic benefits

over in-center hemodialysis treatments, which occur typically bi- or tri-weekly. Studies have shown that more frequent treatments remove more toxins and waste products and render less interdialytic fluid overload than a patient receiving less frequent but perhaps longer treatments. A patient receiving more frequent treatments does not experience as much of a down cycle (swings in fluids and toxins) as does an in-center patient, who has built-up two- or three-days' worth of toxins prior to a treatment. In certain areas, the closest dialysis center can be many miles from the patient's home, causing door-to-door treatment time to consume a large portion of the day. Treatments in centers close to the patient's home may also consume a large portion of the patient's day. HHD can take place overnight or during the day while the patient relaxes, works or is otherwise productive. [0008] Another type of kidney failure therapy is peritoneal dialysis ("PD"), which infuses a dialysis solution, also called dialysis fluid, into a patient's peritoneal chamber via a catheter. The dialysis fluid is in contact with the peritoneal membrane in the patient's peritoneal chamber. Waste, toxins and excess water pass from the patient's bloodstream, through the capillaries in the peritoneal membrane, and into the dialysis fluid due to diffusion and osmosis, i.e., an osmotic gradient occurs across the membrane. An osmotic agent in the PD dialysis fluid provides the osmotic gradient. Used or spent dialysis fluid is drained from the patient, removing waste, toxins and excess water from the patient. This cycle is repeated over a dialysis session, e.g., multiple times.

[0009] There are various types of peritoneal dialysis therapies, including continuous ambulatory peritoneal dialysis ("CAPD"), automated peritoneal dialysis ("APD"), tidal flow dialysis and continuous flow peritoneal dialysis ("CFPD"). CAPD is a manual dialysis treatment. Here, the patient manually connects an implanted catheter to a drain to allow used or spent dialysis fluid to be drained from the peritoneal chamber. The patient then switches fluid communication so that the patient catheter communicates with a bag of fresh dialysis fluid to infuse the fresh dialysis fluid through the catheter and into the patient. The patient disconnects the catheter from the fresh dialysis fluid bag and allows the dialysis fluid to dwell within the peritoneal chamber, wherein the transfer of waste, toxins and excess water takes place. After a dwell period, the patient repeats the manual dialysis procedure, for example, four times per day. Manual peritoneal dialysis requires a significant amount of time and effort from the patient, leaving ample room for improvement.

[0010] Automated peritoneal dialysis ("APD") is similar to CAPD in that the dialysis treatment includes drain, fill and dwell cycles. APD machines, however, perform the cycles automatically, typically while the patient sleeps. APD machines free patients from having to manually perform the treatment cycles and from having to transport supplies during the day. APD machines connect fluidly to an implanted catheter, to a source or bag of fresh dialysis fluid and to a fluid drain. APD machines pump fresh dialysis fluid from a dialysis fluid source, through the catheter and into the patient's peritoneal chamber. APD machines also allow for the dialysis fluid to dwell within the chamber and for the transfer of waste, toxins and excess water to take place. The source may include multiple liters of dialysis fluid including several solution bags.

[0011] APD machines pump used or spent dialysate from the peritoneal chamber, through the catheter, into the drain. As with the manual process, several drain, fill and dwell cycles occur during dialysis. A "last fill" may occur at the end of the APD treatment. The last fill fluid may remain in the peritoneal chamber of the patient until the start of the next treatment or may be manually emptied at some point during the day.

[0012] In any of the above modalities using an automated machine, the automated machine operates typically with a disposable set, which is discarded after a single use. Daily disposables require space for storage, which can become a nuisance for home owners and businesses. Setting up and organizing an APD therapy involves sanitizing work surfaces, un-packaging and positioning various components (e.g., dialysis solution bags and drain bags) about the APD machine, heating dialysis solution bags, connecting bags, etc. Moreover, daily disposable replacement requires daily setup time and effort by the patient or caregiver at home or at a clinic.

[0013] There is also a need for APD devices to be mobile so that the patient may easily transport

the various components for the APD therapy in an organized manner. Additionally, there is a need for APD devices and components to be mobile such that the patient may bring his or her device on vacation or for work travel.

[0014] For each of the above reasons, it is desirable to provide an improved structure and associated functionality for organizing and performing an APD therapy.

SUMMARY

[0015] The present disclosure relates to a multi-function mobile cart to house, organize, and improve the ease of use of an automated peritoneal dialysis (“APD”) system. In general, the APD system may employ an APD machine or cyclor that operates a disposable set.

[0016] In light of the disclosure herein and without limiting the disclosure in any way, in a first aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a mobile dialysis therapy cart includes a top shelf, a cyclor compartment positioned below the top shelf, a fluid bag management compartment, a drain compartment. The cyclor compartment is sized and shaped to house an automated peritoneal dialysis (“APD”) cyclor. The fluid bag management compartment is sized and shaped to house at least two fluid bags. The drain compartment is positioned below the fluid bag management compartment. The mobile dialysis therapy cart may also include a plurality of wheels.

[0017] In a second aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one sanitizing device configured to sanitize at least one location of a disposable set associated with the fluid bags.

[0018] In a third aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the sanitizing device includes a sanitizing light source directed towards the location.

[0019] In a fourth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the sanitizing device includes an ozone generator configured to direct ozone containing gas towards the location.

[0020] In a fifth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the fluid bag management compartment includes a heating device adapted to heat dialysis fluid in one or more fluid bags positioned within the fluid bag management compartment.

[0021] In a sixth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the heating device includes at least one of a resistive plate heater, an inductive heater, and a micro-wave heater.

[0022] In a seventh aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the top shelf includes a hinge configured to enable a portion of the top shelf to rotate upwards to provide access to the cyclor compartment.

[0023] In an eighth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes a drain trolley housed within the drain compartment. Additionally, the drain compartment may include a ramp for the drain trolley.

[0024] In a ninth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the fluid bag management compartment includes at least two telescoping sub-shelves configured to collapse.

[0025] In a tenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one of a urea sensor, a creatinine sensor, a conductivity sensor, a turbidity sensor for sensing patient effluent or a fluid presence sensor for detecting a leak in the fluid bag management compartment.

[0026] In an eleventh aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one

of a pressure sensor, a contact sensor, a strain sensor or a load or weight sensor for sensing a mass or volume of fluid that is delivered to or from the patient.

[0027] In a twelfth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes a level sensor for sensing an orientation of the cart, the cyclor compartment, the fluid bag management compartment, or the drain compartment.

[0028] In a thirteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the drain compartment includes at least one of a pressure sensor, a contact sensor, a strain sensor, a load or weight sensor, a urea sensor, a creatinine sensor, a turbidity sensor or a proximity sensor.

[0029] In a fourteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a mobile dialysis therapy cart includes a top shelf, a fluid bag management compartment, at least one temperature sensor, and a control unit. The fluid bag management compartment is located beneath the top shelf. Additionally, the fluid management compartment is heated and includes at least one heating device for heating a plurality of fluid bags. The at least one temperature sensor is located to sense a temperature associated with dialysis fluid located within the plurality of fluid bags. The control unit is associated with the heating device and the at least one temperature sensor. The control unit is programmed to use an output from the at least one temperature sensor to operate the heating device to heat dialysis fluid within the plurality of fluid bags to a desired temperature.

[0030] In a fifteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes a drain compartment, which may be positioned below the fluid bag management compartment.

[0031] In a sixteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one effluent sensor associated with the drain compartment.

[0032] In a seventeenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes a cyclor compartment positioned below the top shelf. The cyclor compartment may be sized and shaped to house an APD cyclor.

[0033] In an eighteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one sanitizing device. The sanitizing device may be a sanitizing light source associated with at least one of the fluid bag management compartment or the cyclor compartment.

[0034] In a nineteenth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a mobile dialysis therapy cart includes a top shelf, a cyclor compartment, a plurality of fluid bag hooks, and a drain compartment. The mobile dialysis therapy cart may also include a plurality of wheels. The cyclor compartment is positioned below the top shelf. Additionally, the cyclor compartment is sized and shaped to house an APD cyclor.

[0035] In a twentieth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one sanitizing device configured to sanitize a location of at least one fluid bag held by the at least one fluid bag hook.

[0036] In a twenty-first aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the sanitizing device includes a sanitizing light source directed toward the location of the at least one fluid bag.

[0037] In a twenty-second aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the top shelf includes a hinge configured to enable a portion of the top shelf to rotate upwards and fold over itself to provide access to the cyclor compartment.

[0038] In a twenty-third aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes a drain trolley housed within the drain compartment. The drain compartment includes a door, and the door is configured to fold down away from the drain compartment to provide a ramp for the drain trolley.

[0039] In a twenty-fourth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one of a urea sensor, a creatinine sensor, and a turbidity sensor associated with the drain compartment.

[0040] In a twenty-fourth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a mobile dialysis therapy cart includes a top shelf, a cyclor compartment, a fluid bag management compartment, and a drain compartment. The mobile dialysis therapy cart may also include a plurality of wheels. The top shelf includes a top surface. Additionally, the top shelf is adjustable and the top surface is hinged. The cyclor compartment is positioned below the top shelf, and the cyclor compartment is adjustable and is sized and shaped to house an APD cyclor. The fluid bag management compartment includes at least two telescopic sub-shelves. Each of the at least two telescopic sub-shelves are sized and shaped to support a fluid bag. The drain compartment is positioned below the fluid bag management compartment. Additionally, the drain compartment includes a door that is adapted to open from the top and fold down towards a bottom of the dialysis therapy cart. The door may serve as a ramp surface for a drain trolley housed in the drain compartment.

[0041] In a twenty-sixth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes a display device that is removable from the cart while maintaining connectivity with the cart.

[0042] In a twenty-seventh aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the mobile dialysis therapy cart includes at least one of a urea sensor, a creatinine sensor, a conductivity sensor, and a turbidity sensor associated with the drain compartment.

[0043] In a twenty-eighth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, a dialysis therapy system includes a cart, an APD cyclor, and at least two fluid bags. The cart includes a top shelf, a cyclor compartment positioned below the top shelf, a fluid bag management compartment, and a plurality of wheels. The APD cyclor is housed within the cyclor compartment, which is sized and shaped to house the APD cyclor. The at least two fluid bags are housed within the fluid bag management compartment, which is sized and shaped to house at least two fluid bags.

[0044] In a twenty-ninth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the cart includes a drain compartment positioned below the fluid bag management compartment.

[0045] In a thirtieth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the dialysis therapy system includes a drain bag. The drain bag may be housed within the drain compartment.

[0046] In a thirty-first aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the dialysis therapy system includes at least one of a urea sensor, a creatinine sensor, and a turbidity sensor associated with the drain bag or the drain compartment.

[0047] In a thirty-second aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the dialysis therapy system includes a sanitizing device configured to sanitize a portion of the dialysis therapy system.

[0048] In a thirty-third aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the sanitizing device is at least one of a sanitizing light source or an ozone generator.

[0049] In a thirty-fourth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, the dialysis therapy system includes at least one of a load or weight sensor, a pressure sensor, a strain sensor, and a level sensor.

[0050] In a thirty-fifth aspect of the present disclosure, which may be combined with any other aspect listed herein unless specified otherwise, any of the features, functionality and alternatives described in connection with any one or more of FIGS. 1 to 4 may be combined with any of the features, functionality and alternatives described in connection with any other of FIGS. 1 to 4.

[0051] It is accordingly an advantage of the present disclosure to provide a mobile cart with a compact footprint to reduce the space required to conduct an APD treatment in a patient's room.

[0052] It is another advantage of the present disclosure to provide a mobile cart that allows a patient to move APD therapy equipment to different rooms and locations and to improve patient mobility during treatment.

[0053] It is yet another advantage of the present disclosure to provide a multi-function cart that organizes components for APD therapy.

[0054] It is a further advantage of the present disclosure to provide a cart capable of heating and maintaining the temperature of multiple dialysis solution bags.

[0055] It is yet a further advantage of the present disclosure to provide an easily removable and mobile drain bag trolley for final wasting of effluent fluid.

[0056] It is yet another advantage of the present disclosure to provide a multi-function cart that is configurable to enable optimal use and organization for varied bedroom environments.

[0057] It is a further advantage of the present disclosure to provide a cart that facilitates a repeatable set-up procedure with common locations for therapy components to reduce error and optimize instrument operation.

[0058] Additional features and advantages are described in, and will be apparent from, the following Detailed Description and the Figures. The features and advantages described herein are not all-inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the figures and description. Also, any particular embodiment does not have to have all of the advantages listed herein and it is expressly contemplated to claim individual advantageous embodiments separately. Moreover, it should be noted that the language used in the specification has been selected principally for readability and instructional purposes, and not to limit the scope of the inventive subject matter.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0059] FIG. 1 is a perspective view of an automated peritoneal dialysis (“APD”) machine operating with a disposable, fluid carrying set.

[0060] FIGS. 2A, 2B and 2C are perspective views of a first embodiment for a mobile cart of the present disclosure.

[0061] FIG. 3A is a perspective view of a second embodiment for a mobile cart of the present disclosure.

[0062] FIG. 3B is a front elevation view of the second embodiment for the mobile cart of the present disclosure.

[0063] FIG. 3C is a rear elevation view of the second embodiment for the mobile cart of the present disclosure.

[0064] FIG. 3D is a side elevation view of the second embodiment for the mobile cart of the present disclosure.

[0065] FIGS. 3E and 3F are top perspective views of the second embodiment for the mobile cart of the present disclosure.

[0066] FIG. 4 is a schematic diagram of a mobile cart of the present disclosure.

DETAILED DESCRIPTION

[0067] Referring now to the drawings and in particular to FIG. 1, a typical configuration **100** for an automated peritoneal dialysis (“APD”) system **102** includes an APD machine or cyclor **104** that operates with a disposable set **118** including a disposable cassette (located within cyclor **104**), fluid containers or bags **120a** to **120c** and associated lines or tubing **122**. The fluid containers or bags **120a** to **120c** may be dialysis fluid bags, dialysis solution bags, heating bags, solution bags, concentrate bags, etc. and may be commonly and generally referred to herein as fluid bags **120**. APD machine or cyclor **104** in the illustrated embodiment includes a heating tray **106** atop the APD machine or cyclor **104**. During setup regarding cyclor **104** and disposable set **118**, a patient may implement “ad hoc” solutions to organize and position the various components required to perform an APD therapy. For example, the patient typically has to find space to unpack the various fluid bags **120a** to **120c** and lines **122** and place them somewhere near APD machine **104**.

[0068] Specifically, a patient may place a fluid bag **120a** (e.g., dialysis solution or heating bag) on the heater or heating tray **106**. Then, the patient may position other fluid bags **120b, c** (e.g., dialysis solution bags) near the APD machine or cyclor **104** for connection to the disposable cassette. For example, each evening, the patient may set up the APD machine or cyclor **104** to perform three to five fluid treatment exchanges. To perform three to five exchanges, the patient may connect three to five fluid bags **120a** to **c** (e.g., dialysis solution bags) to the disposable cassette operating with the cyclor **104**. A heating source, such as the cyclor's **104** heating tray **106** may warm the fresh dialysis solution before it enters the patient's body. Additionally, the patient may position a drain bag **140** near the APD machine or cyclor **104** to check drain fluid turbidity.

[0069] Implementing “ad hoc” solutions to organize and position the various components required to perform the APD therapy creates stress and discomfort and poses a sanitary risk for patients. For example, a patient's room may provide inadequate space for preparation and organization of a therapy treatment. Additionally, the portions of the patient's room that are used as an “ad hoc” solution should be manually cleaned each day prior to treatment setup, which is burdensome to the patient. Improper loading of the fluid bag **120a** (e.g., dialysis solution or heating bag), for example, may lead to only portions of the dialysis fluid volume being heated at a time, causing delays and time loss waiting for dialysis solution to warm to an appropriate temperature before entering the patient's body.

[0070] In addition, a PD patient typically has lower mobility and flexibility due to the mobility constraints created by the “ad hoc” solutions. For example, the repositioning of the cyclor **104** may be limited by the length of the cyclor's **104** power cord and/or by the physical location of the cyclor **104** and of the fluid bags **120** (e.g., dialysis solution bags and drain bag **140** placed onto small furniture such as a small chair, table or nightstand in the patient's bedroom). Even though each of the components for an APD therapy is independently portable (e.g., the cyclor **104**, fluid bags **120** and drain bag **140**), the components together drastically constrain movement. The patient's traveling capabilities with “ad hoc” solutions are also limited due to the combined mobility constraints created by the components, for example, which may require cleaning of the various surfaces at the new location, lifting and carrying each component (e.g., cyclor **104**, fluid bags **120**, drain bag **140**, etc.) to the new location, etc.

[0071] FIGS. 2A to 2C illustrate an APD therapy management system **200** that includes a cart **210**, which allows the system **200** to be moved readily, e.g., from a family room to a bedroom and vice versa. The cart **210** may be a mobile cart and/or a multi-function cart, and/or a medical therapy cart, such as a dialysis therapy cart. For example, cart **210** may include wheels **280a** to **280d** that allow cart **210** to be moved from one room to another in the patient's house. During cleaning, cart **210** may be moved to a kitchen or utility room in which cleaning supplies are readily accessible. Additionally, the mobility of cart **210** allows the cart **210** to be rotated or positioned in a preferred direction of operation (e.g., facing patient's bed). One or more handle (not pictured) may be

provided to facilitate the movement of the cart **210**. In one embodiment, the handle may be rotated out of the way (e.g., upwardly or downwardly) when not needed.

[0072] Wheels **280a** to **280d**, hereinafter referred to generally as wheels **280**, may be casters or rollers that enable cart **210** to be pushed along flat surfaces. Additionally, wheels **280** may be sized such that cart **210** can move along carpet and over other small obstacles. Each of the wheels **280** may rotate independently of the others, and around both horizontal and vertical axis, to provide improved turning and cart mobility in tight spaces. Cart **210** may include three, four, or more wheels **280**. Additionally, wheels **280** may include stops or brakes to prevent or limit further movement once the cart **210** is moved to a desired location.

[0073] Cart **210** may be powered via AC power and/or battery power. For example, cart **210** may include a battery (e.g., battery **415** of FIG. 4), which is used to provide the APD machine or cyclor **104** with battery power when no AC power is available. For example, to move system **200**, the patient may unplug a power cord and proceed to wheel cart **210** to any desired destination. Battery **415** may be a long-lasting battery powering the cyclor **104**, while allowing cart **210** to move to another room and avoiding the discontinuation of treatment while the patient moves. In an embodiment, battery **415** is a rechargeable battery, which is recharged while cart **210** is connected to AC power.

[0074] FIGS. 2A to 2C illustrate that cart **210** may include a top shelf **220**, a cyclor compartment **230**, a fluid bag management compartment **240**, and a drain compartment **250**. Compartments **230**, **240**, **250** may be enclosures similar to a drawer or a lock-box with a door. In another example, compartments **230**, **240**, **250** include shelves that are open on at least one side. Compartments or shelves **220**, **230**, **240** and **250** may hold supplies needed for a dialysis therapy and organize each of the therapy components. For example, the compartments and shelves enable a patient to organize and route tubing from fluid bags **120** in fluid bag management compartment **240** that extend to the disposable cassette interfacing with the cyclor **104** located in the cyclor compartment **230**. A patient tube leads from the disposable cassette in the cyclor compartment **230** to the patient. A drain line leads from the disposable cassette in the cyclor compartment **230** to the drain bag **140** located in the drain compartment **250**.

[0075] Top shelf **220** and compartments **230**, **240**, **250** or shelves provide surfaces having enough stability to support therapy components housed in cart **210**, which further protects the therapy components against accidental bumps and contacts. In an example, the compartments or shelves may be open from the front side of cart **210**. FIG. 2A illustrates that one or more of the compartments or shelves **230**, **240**, **250** may include a respective door **232**, **242**, **252**, which may be hinged doors.

[0076] Door(s) **232**, **242**, **252** may include respective locking mechanisms. The locking mechanisms maintain doors **232**, **242**, **252** in a closed position allowing the cart **210** to be transported and moved without any of the APD components falling out of the cart. The locking mechanisms may additionally securely store APD components and keep those components out of reach from children, pets, etc.

[0077] FIG. 2B illustrates that fluid bag management compartment **240** includes multiple slots or sub-shelves (e.g., sub-shelves **244a** to **244c**) that organize fluid bags **120** at the beginning of the treatment. In the illustrated example, fluid bag management compartment **240** includes three slots or sub-shelves **244a** to **244c** for three fluid bags **120** that serve to supply additional dialysis fluid for future treatment cycles. Additionally, another fluid bag **120a** (e.g., heating bag) sits atop the cyclor **104** in one embodiment. In other examples, fluid bag management compartment **240** may be configured to accommodate additional fluid bags **120**, while fluid may take place without requiring the patient to place a bag or container atop the cyclor **104**.

[0078] In an example, each of the slots or sub-shelves **244a** to **244c** is telescopic or otherwise moveable, such that each shelf can collapse or move inward and out of the way when not used to support a fluid bag **120**. The slots or sub-shelves **244a** to **244c** may alternatively or additionally be

hinged and capable of folding out of the way (e.g., folding upwardly) to provide additional access to the shelf below. For example, the slot or sub-shelf **244a** may telescopically collapse inward and/or fold upward and out of the way to provide additional access to the slot or sub-shelf **244b**. In other examples, the slots or sub-shelves **244a** to **244c** may be telescoping, collapsed or folded out of the way to accommodate larger fluid bags. The slots or sub-shelves **244a** to **244c** may further alternatively or additionally be removable and placed for example on the bottom of cart **210**.

[0079] Still further alternatively or additionally, fluid bag management compartment **240** and its associated slots or sub-shelves **244a** to **244c** may be sloped to orient each fluid bag so that the bag outlet port resides below the rest of the bag for optimum drainage and air handling. The sloped configuration of fluid bag management compartment **240** advantageously causes dialysis fluid to flow from the bags until emptied, leaving air free to migrate up towards the back of the bags and to remain trapped therein. Similarly, drain compartment **250** may also be sloped to assist with flow into or out of drain bag **150**.

[0080] As shown specifically in FIG. 2B, APD machine or cyclor **104** can, for example, be located within cyclor compartment or shelf **230** above fluid bag management compartment or shelf **240**, which provides ready access to the slots or subshelve(s) **244a** to **244c**.

[0081] The cart **210** may also include a display device **290**. The display device **290** may be integrated with or be a removable or separable from the cart **210**. The display device **290** may be communicatively coupled to the cyclor **104** and/or any of the sensors as discussed in more detail below for the presentation of information to the patient. For example, the display device **290** may present user interface information from the cyclor **104** to the patient. The display device **290** may advantageously present information to the patient when the input devices and the display of cyclor **104** is facing away from the patient. For example, a patient may be unable to position the cart **210** such that the input devices and display of the cyclor **104** face the patient's bed. However, the patient may remove and reposition the display device **290** in a convenient location that is viewable by the patient even when the cyclor **104** is out of sight.

[0082] As illustrated in FIG. 2C, the cart **210** accommodates drain bag **140**, e.g., in drain compartment **250**. The cart **210** may also include a drain trolley **260** for draining and transporting drain bag **140**. The drain trolley **260** may be self-contained. If a drain line is run to a house drain instead of a drain bag **140**, the drain line may be removed from the drain and placed in the drain compartment **250** (e.g., within drain trolley **260**) when system **200** is moved. The drain trolley **260** allows for easy transport of drain bag **140** to a restroom for a final draining of effluent waste fluid, e.g., to a toilet or bathtub. The drain trolley **260** advantageously allows a patient to transport drain bag **140** without having to move the entire cart **210** into the restroom. In an example, a door **252** additionally serves as a ramp for trolley **260** to slide down from the bottom of the compartment to the floor or ground. The door **252** (along with other doors described herein) may be a front door. In other examples, a long drain tube may be incorporated into the drain compartment **250**, which enables waste fluid to be drained without having to transport or empty a heavy drain bag **140**.

[0083] Alternatively, cart **210** may include a drain line instead of a drain compartment **250** or drain trolley **260**. For example, cart **210** may be positioned to drain fluid directly via a drain line without temporarily storing the fluid in drain compartment **250** or drain trolley **260**. The drain line may be of sufficient length to allow a patient to directly drain effluent waste fluid in a toilet or a bathtub.

[0084] FIGS. 3A to 3F illustrate an alternate APD therapy management system **300**, which includes alternative cart **310**. The cart **310** may likewise be a mobile cart and/or a multi-function cart, and/or a medical therapy cart, such as a dialysis therapy cart. Cart **310** in the illustrated embodiment includes a top shelf **320**, a cyclor compartment or shelf **330**, and a drain compartment or shelf **350**. Similar to FIGS. 2A to 2C, compartments or shelves **330** and **350** may be or form compartments. Instead of placing the supply bags in a fluid bag management compartment or shelf **240** (illustrated in FIGS. 2A to 2C), the cart **310** in the illustrated embodiment includes a plurality of fluid bag fingers or hooks **340a** to **340n**, hereinafter referred to generally as fluid bag fingers or hooks **340** or

simply as fingers or hooks **340**. For example, a finger or hook **340** may support a fluid bag and may be described as a fluid bag hook; or the hook **340** may support a drain bag and may be described as a drain bag hook. In the embodiment illustrated in FIGS. **3A** to **3D**, the compartments or shelves are open from each side of the cart **310**. The cart **310** is therefore configured to allow a patient or a caregiver to place hooks **340a** to **340n** along any one or more desired side of the cart **310**.

[0085] Similar to cart **210**, cart **310** may include wheels **380a** to **380d**, which allow cart **310** to be moved from one room to another in the patient's house or to be repositioned in a preferred orientation for operation (e.g., facing patient's bed). Positioning cart **210** or **310** in a preferred direction or orientation, such as facing the patient's bed, allows the input devices and the display of the cyclor **104** to face the patient's bed while the patient falls asleep. Here, if an alarm sounds, the patient can access the cyclor **104**, the patient line, the fluid bags **120** and/or the associated lines, etc., without having to get out of bed. A handle (not pictured) may facilitate moving cart **310**. In one embodiment, the handle may be rotated out of the way (e.g., upwardly or downwardly) when not needed. The patient may use the handle or otherwise turn cart **310** during treatment, e.g., while in bed, to access any desired area of the cart **310**.

[0086] Wheels **380a** to **380d**, hereinafter referred to generally as wheels **380**, may be casters or rollers that enable the cart **310** to be pushed along flat surfaces. Additionally, wheels **380** may be sized such that cart **310** may easily move along carpet and other small obstacles. Each of the wheels **380** may rotate independently of the others, and around both horizontal and vertical axis, to provide improved turning and cart mobility in tight spaces. Cart **310** may include three, four, or more wheels **380**. Additionally, wheels **380** may include stops or brakes **382** to prevent or limit further movement once the cart **310** is in a desired location.

[0087] Fingers or hooks **340** may be placed in various positions and configurations on the cart **310**. As illustrated in FIG. **3B**, the cart **310** may include rails **344**. The rails **344** may provide mounting points for fingers or hooks **340**. In an example, rails **344** may be adjustable such that the rails can be moved to different heights between compartments or shelves **320**, **330** and **350**. Rails **344** and thus fingers or hooks **340** are therefore capable of being placed on any side of the cart **310** to support fluid bags **120**. Fluid bags **120** hanging from fingers or hooks **340** are oriented such that the bag outlet ports reside below the rest of the bag for optimum drainage and air handling, which advantageously causes dialysis fluid to flow from the fluid bags **120** until emptied, leaving air free to migrate up towards the top of the bags and to remain trapped therein.

[0088] As discussed above, fluid bags **120a** to **120c** (e.g., dialysis solution bags **120**) may be suspended on fingers or hooks **340** arranged about desired sides of the cart **310**. For example, fingers or hooks **340** may be positioned on the side of the cart **310** as illustrated in FIG. **3A** and/or along the front of the cart **310** as illustrated in FIG. **3B**. In another example, fingers or hooks **340** are positioned along the top portion of a fluid bag management compartment (e.g., similar to fluid bag management compartment **240** of FIGS. **2A** to **2C**), which is sized and shaped to allow fluid bags **120** to hang within the compartment while also being contained within the cart **310**.

Alternatively or additionally, a dialysis solution or fluid bag **120** may sit inside a shallow concave depression forming the heating tray **106** atop the APD machine or cyclor **104**, which may be sized and shaped to accommodate a typical 5 liter bag of PD solution, for example.

[0089] Similar to cart **210**, the various compartments, shelves, fingers or hooks of the cart **310** may hold any ancillary supplies needed for a dialysis therapy and to organize each of the supplies and therapy components. Carts **210** and **310** not only provide a means to optimally organize treatment, carts **210** and **310** also allow the patient to setup for a next treatment when tearing down a just-completed treatment, e.g., in the morning. In combination with the pre-setup, systems **200** and **300** allow the patient to pre-program a start time for the next treatment, so that the system **200** or **300** is fully primed and ready to commence in the day or evening.

[0090] As illustrated in FIGS. **3E** and **3F**, top shelf **320** of cart **310** (and similarly top shelf **220** of cart **210**) provides a work surface for the patient to initially organize and connect fluid bags. In

some instances, cycler **104** may be positioned on top shelf **320**. However, by providing a cycler compartment **230** (see FIGS. **2A** to **2C**) or a cycler compartment or shelf **330**, the top shelf **320** is advantageously available as a work surface. Top shelf **320** in the illustrated embodiment includes a hinge **390** such that top shelf **320** may fold to open upwardly to provide direct access to heating tray **106**. For example, hinge **390** on top shelf **320** may form a hinged top surface. FIG. **3G** for example illustrates top shelf **320** in a folded configuration in which half of shelf **320** is folded about hinge **390** so as to lay upon the other half of the shelf. Doing so provides additional access to heating tray **106** and cycler **104** in general as well as to fluid bags **120a** (e.g., dialysis solution and/or heating bag) located on heating tray **106**. Hinge **390** is accordingly located along the middle [0091] As illustrated in FIGS. **3A**, **3B** and **3D**, cycler compartment or shelf **330** may include a protruding portion **334** that extends beyond other shelves **320**, **340** to allow space for tube routing from the cycler **104** (e.g., cassette) to various solution bags arranged on hooks **340**. Additionally, protruding portion **334** may allow the cycler **104** to sit closer to the edge of the cart **310** and provide easier access and visibility to the controls and display of the cycler **104**.

[0092] Similar to hooks **340**, each compartment or shelf **330**, **350** may also be adjustable to accommodate different types and models of cycler **104**, different sizes and styles of fluid bags, and different drain configurations. Additionally, the overall height of carts **210** and **310** may also be adjustable. For example, a patient may adjust the height of cart **210**, **310** based on the patient's bed height. Adjustment may be via the extension and contraction of the vertical posts of the cart **310**, via adjustments made at wheels **280a** to **280d** or wheels **380a** to **380d**, and/or via adjustment to the height of Compartments **230**, **240**, **250** for cart **210**.

[0093] Similar to hooks **340**, cart **310** (and cart **210**) may also include a holster (not shown) to hold hand sanitizer. Providing easy access to hand sanitizer minimizes the risk of contaminating the various components required to perform an APD therapy. For example, a patient may use the hand sanitizer prior to connecting solution bags to control possible bacterial contamination introduced while connecting solution bags. The sanitizer holster may be positioned in different locations about carts **210**, **310** similar to hooks **340**, which provides an easily accessible location for hand sanitizer, and wherein the patient does not need to open or close any of the compartments for access to the hand sanitizer.

[0094] Other holsters or compartments may additionally be positioned about carts **210**, **310** for other components (e.g., consumables) such as gloves, caps, gauze, masks, tape, waste, etc.

[0095] Overall, the construction of carts **210**, **310** is highly configurable to enable optimal use in a wide variety of different bedroom environments. In addition to the overall height of the carts **210**, **310**, the relative position of each compartment or shelf may also be adjustable. For example, the position of each shelf may be adjustable to accommodate different types and sizes of components and cyclers **104**. Additionally, as discussed above, shelf height or overall cart height is adjustable in various embodiment to improve ease of use for patients of different heights and to adapt the cart according to the patient's bed height.

[0096] FIG. **4** is a schematic view of another cart **410** of the present disclosure. It should be appreciated that all components, features, functionality and alternatives discussed in connection with carts **210** and **310** is also applicable to cart **410** and vice versa. For example, a feature described in relation to cart **310** (e.g., hooks **340**) is similarly applicable to and/or combinable with carts **210**, **410**, such that those carts also embody that feature. The same applies for any feature of carts **210** and **410** being applicable to and/or combinable with the other carts. It should also be appreciated that different components of each of carts **210**, **310** and **410** may be made of any combination of metal, plastic or wood. If metal, the component may for example be aluminum, steel or stainless steel. If plastic, the component may for example be polyvinyl chloride ("PVC"), polyethylene ("PE"), polyurethane ("PU") and/or polycarbonate.

[0097] As illustrated in FIG. **4**, system **400** includes a cart **410**, which allows the system **400** to be moved readily from one location to another. The cart **410** may be a mobile cart, and/or a multi-

function cart, and/or a medical therapy cart, such as a dialysis therapy cart. The cart **410** includes a top shelf **420**, a cyclor compartment **430**, a fluid bag management compartment **440**, and a drain compartment **450**. Top shelf **420**, cyclor compartment **430**, fluid bag management compartment **440**, and drain compartment **450** may include the same or similar features as the respective compartments and shelves described in connection with the carts **210**, **310**.

[0098] The cart **410** in the illustrated embodiment includes a battery or power supply **415**, a heater or heating compartment **460**, sanitizing, e.g., ultraviolet (“UV”) light source(s) **470a** to **470d** configured and positioned and arranged to disinfect desired surfaces and chambers or cart **410**. The cart **410** also includes sensor module **480**, which may include one or more temperature sensor(s) **482**, urea sensor(s) **484**, creatinine sensor(s) **486**, conductivity sensor(s) **499** and fluid presence sensor(s) **488**. Temperature sensor(s) **482** are used in connection with onboard heating, while urea sensor(s) **484**, creatinine sensor(s) **486**, conductivity sensor(s) **499** and fluid presence sensor(s) **488** are used in connection with dialysis fluid sensing. For example, the conductivity sensor(s) **499** may sodium concentration information measured by the conductivity sensor(s) **499**, which may provide information related to clearance. In an example, the conductivity sensor(s) **499** may be a sodium concentration sensor.

[0099] Cart **410** may implement shelves, compartments, trays, surfaces or hooks. For example, instead of a cyclor compartment **230**, cart **410** may instead include a cyclor shelf. Furthermore, instead of fluid bag management compartment **240**, cart **410** may instead include a fluid bag management shelf, a fluid bag management tray, or fluid bag hooks. Additionally, top shelf **220** may instead be a top surface, a top tray or a top compartment. In another example, instead of drain compartment **450**, cart **410** may include a drain shelf or a drain hook to hold the drain bag **140**. It should be appreciated that for each of the carts **210**, **310**, **410** discussed herein, each of the support structures may be in the form of shelves, compartments, trays, surfaces or hooks.

[0100] Any of the carts **210**, **310** and **410** may include a handle (not pictured) to assist a patient in pushing or pulling the cart. As discussed in more detail below, the cart **210**, **310**, **410** may be configured to have an auto-adjustable height, shelf height, or compartment height to improve the layout of the cart for a specific application. For example, the overall cart height and various shelf or compartment heights may be adjusted based on a patient's bed height, a patient's reach, or a patient's height to provide optimum functionality including improved sightlines and instrument positioning. Additionally, the auto-adjustment or auto-leveling may ensure that the shelves or compartments provide a substantially horizontal support surface during use. The auto-adjustment or auto-leveling may be performed by leveling screws and an electric motor powered by a battery (e.g., onboard battery) or power supply **415**.

[0101] The battery or power supply **415** may also provide power-assist to the wheels when the patient is pulling or pushing the cart. For example, the handle (not pictured) may include a power-assist button or switch, similar to that of a self-propelled vacuum or self-propelled electric lawn mower, which the patient may actuate to provide power to rotate the wheels while moving the cart.

Onboard Heater

[0102] It should be appreciated that carts **210**, **310** and **410** are not limited to storing and operating with any particular type of cyclor, including cyclor **104**. To that end, it is not required that the cyclor operating with carts **210**, **310** and **410** provide heating tray **106**. It is contemplated to instead provide a more compact and simplified cyclor, which does not heat the dialysis fluid. In such a situation, and even in the case that the cyclor provides a heater, it is contemplated that any of carts **210**, **310**, **410** (here illustrated with cart **410**) provide a heater **460** (e.g., onboard heater), such as a heating compartment to heat fluid bags. In various examples, heater **460** is implemented as an entire compartment or shelf dedicated to heating of the fluid bags, such as dialysis solution bags **120**. Heater **460** is associated with the fluid bag management compartment or shelf. By dedicating a compartment or shelf for heating fluid bags **120** (e.g., dialysis solution bags), each of the fluid bags **120** can be efficiently heated at the same time. Additionally, the heating portion of carts **210**,

310, 410 may keep each of the fluid bags **120** warm such that they are constantly ready for use. [0103] In an example, heater **460** may be a thermostatic device integrated in one or more of a rear panel, a side panel, a top or a bottom of the fluid bag management compartment or shelves of the cart **210, 310, 410**. In various examples, the heater **460** may be a resistive plate heater or include multiple resistive heaters, e.g., a resistive heater for each fluid bag **120**. In an alternative embodiment, carts **210, 310, 410** may be equipped with a heating blanket that can be placed around fluid bags. Additionally or alternatively, a heating sleeve may be placed over one or more of the solution bags to heat the solution while the one or more bag sits in fluid bag management compartment or shelves or hangs from hooks **340**. Radiant heating may also be used alone or in addition to resistive heating. In other examples, the heater **460** may be adapted for inductive heating, micro-wave heating or the like.

[0104] Incorporating a thermostatic device within carts **210, 310, 410** allows multiple fluid bags **120** to be heated and kept warm at the same time, which avoids the need to lift a fluid bag **120a** (e.g., heating bag) onto the heater tray **106**. Both heating and maintaining the heat of multiple fluid bags **120** also reduce set-up time between treatments. To further reduce set-up for treatment time, the treatment time may be further optimized by heating the fluid bags **120** prior to the treatment by means of a timer function. The heater **460**, such as a thermostatic device, may include a plurality of heating coils that are embedded within the compartment or below the shelf surface. Carts **210, 310, 410** in the illustrated embodiment include one or more temperature sensor **482** positioned on or within the heated surface to track the temperature of the solution in the fluid bags **120**. For example, temperature sensor(s) **482** may provide feedback to a control unit **462** for heater **460**, which includes a control circuit that turns heater **460** on and off as needed to maintain the dialysis fluid at the desired temperature.

[0105] In an embodiment, fluid bag management compartment having onboard heater **460** is insulated, e.g., includes insulated sides, rear panel, top and bottom and perhaps the door. The insulation increases the efficiency of the fluid bag management compartment that is heated by heater **460**. Each of the heaters **460** described herein, such as the heating tray **106**, heating compartment, heating shelf, onboard heater, resistive plate heater, heating blanket, heating sleeve, infrared heater, inductive heater, micro-wave heater etc., may be referred to generally as a heating device **460**.

Light Sanitization

[0106] Carts **210, 310, 410** may include one or more sanitizing device **468a** to **468c** configured to sanitize shelves, compartments, or components housed on or within the carts, such as the sanitizing of specific location(s) of a disposable set associated with the fluid bags. The sanitizing devices **468a** to **468c** may be generally referred to as sanitizing device(s) **468**. The sanitizing devices **468** may be light sources **470a** to **470d** that emit light capable of sterilization or disinfection. In addition to the holster providing hand sanitizer, light source(s) **470a** to **470d**, hereinafter referred to generally as light source(s) **470**, are configured to emit sanitizing, e.g., UV light capable of sterilization or disinfection to further control possible bacterial contaminations introduced to the various therapy components (e.g., when connection solution bags). For example, light source(s) **470** may be positioned and arranged to illuminate the solution bag connectors in the fluid bag management compartments **240, 440** and a similar compartment in cart **310**.

[0107] Light source **470** may be an ultraviolet (“UV”) light source, such as UV bactericidal lights. In other examples, the light source **470** may be a mercury vapor lamp, a xenon flash lamp, a continuous arc lamp, and may include multiple UV light emitting diodes (LEDs), etc. Light source **470** may emit UVA, UVB, or UVC light. Additionally, light source **470** may emit both UV and visible light. By emitting UV light from different portions of the electromagnetic spectrum, carts **210, 310** and **410** may be more effective at sanitizing APD components over a broader range of microorganisms.

[0108] Light may be emitted from light sources **470** so as to provide continuous sterilizing

radiation over a period of time. Alternatively, light is emitted in pulses or flashes at selected times and at predetermined intervals. In an example, the energy of a single pulse may be sufficient to deliver a sanitizing or disinfecting dosage to the APD components within the compartment, e.g., connector locations or areas of fluid bags **120**. To that end, light sources **470** are positioned and aimed to direct sanitizing light towards the desired locations and areas of the solution bags or other desired location of disposable set **118**. The light sources **470** may for example be mounted to the sides, tops or rear surfaces of carts **210**, **310** or **410** and/or to a frame of the carts.

[0109] Each compartment or shelf may be associated with its own light source **470** (e.g., disinfecting light source or sanitizing light source). In the example illustrated in FIG. **4**, cyclor compartment **430** is associated with light source **470a**, fluid bag management compartment **440** is associated with light source **470b**, while drain compartment **450** is associated with light source **470c**. Additionally, top shelf **420** may be associated with light source **470d**. Any of light sources **470a** to **470d** may be a light provided on the end of a flexible “snake” type connection, or a universal hinge type connection, which can be pointed at different regions of, e.g., the top shelf **420**, to provide “spot” sanitization. The intensity, treatment period, quantity or frequency of pulses, etc., of the light emitted by light sources **470** may be adjusted for each compartment or shelf. For example, the compartments or shelves may house components made of different materials that may require different treatment parameters for sterilization or disinfection. Additionally, the compartments or shelves along with the components housed within those compartments or shelves may be designed such that the sanitizing light (e.g., UV light) does not damage or deteriorate the compartments, shelves or components housed within the compartments or shelves. For example, the compartments, shelves and/or components may be made from UV-resistant materials.

[0110] The light sources, which may emit sanitizing light such as UV light, advantageously help to maintain sterility and reduce the threat of contamination to the patient. Sanitizing components via light source(s) **470** may reduce the risk of peritonitis. Each shelf, compartment, tray or hook may be protected with light opaque materials that minimize UV light or other light from exiting the enclosure, thereby minimizing the exposure to patients and other bystanders. Therefore, the patient may safely and comfortably use cart(s) **210**, **320**, **410** without unwanted exposure to UV light or other light, especially during night use.

Ozone Sanitization

[0111] The carts **210**, **310**, **410** may alternatively or additionally perform sanitization and/or sterilization using ozone containing gas. For example, the carts may include one or more sanitizing devices **468** configured to sanitize shelves, compartments, or components housed on or within the carts, such as sanitizing specific location(s) of a disposable set associated with the fluid bags. The sanitizing device **468** configured to sanitize may include an ozone generator. The sanitizing device **468** may also include other components associated with ozone generator, which are discussed in more detail below. The ozone may be produced externally and then supplied to one or more compartments of the carts **210**, **310**, **410**. In an example, medical quality oxygen is converted into ozone containing gas in an ozone generator. The oxygen may be supplied from a supply tank. Additionally, the ozone containing gas may be stored in a storage tank. The ozone generator may be housed on or housed within the cart **210**, **310**, **410**. Similarly, the oxygen supply tank and storage tank may be housed on or within the cart. In another example, the ozone generator, oxygen supply tank, and storage tank may be located external to the cart. The cart **210**, **310**, **410** may also include a vacuum pump associated with the compartments to provide a sufficient vacuum to the compartment to increase the penetration of the ozone gas.

[0112] The medical quality oxygen may pass through a filter before being converted into ozone containing gas. Additionally, the ozone containing gas may pass from the supply tank, through a filter, a pressure regulator, a flow meter and/or shut-off valve before entering one of the cart's compartments. For example, the ozone containing gas may be directed to one of the cart's compartments through a flow regulator or pressure regulator. The carts **210**, **310**, **410** and any

associated compartments may include covers or seals (e.g., gasket seals) for each compartment to ensure the ozone containing gas is retained in a target compartment to perform sanitization or sterilization.

Sensors

[0113] Carts **210, 310, 410** may include a desired sensor module **480**. As discussed above, carts **210, 310, 410** may include temperature sensors **482** positioned on or within the compartments having onboard heated surfaces to track the temperature of solution within fluid bags **120**. As discussed above, temperature sensor(s) **482** may be used as feedback to a control unit **462** of heater or heating device **460** (e.g., onboard heater), which turns heater **460** on and off as needed to maintain dialysis fluid at a desired temperature, e.g., body temperature of 37° C.

[0114] FIG. 4 illustrates that urea, creatinine and/or turbidity sensors **484, 486** may be employed so as to sense the patient's drain fluid or effluent, e.g., be incorporated into the drain bag **140** or along its associated drain line to sense one or more fluid properties of the effluent in the drain bag **140**. Additionally, conductivity sensors **499** may provide conductivity information or sodium concentration information of the fluid. For example, the sensors may be used to identify substances (e.g., urea and creatinine) as well as the concentration of such substances, which provides information regarding the composition of the solution in the drain bag **140**. The sensors may alternatively or additionally look for white cells as an indicator of peritonitis. Sensors **484, 486** and/or **488** may be optical and/or inductive or capacitive sensors. Other sensors (not pictured) may include an effluent sensor, however in some examples, the effluent sensor may generally refer to any one of the urea sensors **484**, creatinine sensors **486** or turbidity sensors.

[0115] Each of the compartments, shelves, surfaces or trays of carts **210, 310** and **410** may also include leakage sensors or fluid presence sensors **488**. For example, the leakage sensors or fluid presence sensors **488**, e.g., electrical contacts, may detect the presence of a fluid's leakage within a compartment or on a shelf surface (e.g., fluid that is leaking from a bag or a connection line). Leakage sensors or fluid presence sensors **488** advantageously provide early detection and warnings to the patient if a fluid bag is leaking or if a bag connector is not properly attached. Sensors **484, 486** and **488**, in an embodiment, output to a control unit, such as the same control unit **462** for the heater **460**. The sensors may provide system and patient data that can be delivered to a desired location and/or output to an alarm, e.g., audio, visual or audiovisual alarm.

[0116] Additionally, each of the compartments, shelves, surfaces, trays or hook of carts **210, 310** and **410** may also include a load or weight sensor(s) **490**, a pressure sensor(s) **491**, a strain sensor(s) **492**, a level sensor(s) **493**, or any combination thereof. For example, the outputs of any of the load or weight sensors **490**, pressure sensors **491**, strain sensors **492** or level sensors **493** may be used to provide volume or flowrate monitoring and control both to and from the patient. The sensors may measure an infused mass, an infused volume, a drained mass, or a drained volume to or from the patient. Measurements may be obtained in compartments, shelves, surfaces, trays in which sanitization or disinfection is taking place (e.g., sanitizing connectors). Additionally, measurements may be obtained in compartments, shelves, surfaces, trays in which heating is taking place (e.g., heating fluid bags **120**). In an example, the load or weight sensor(s) **490**, pressure sensor(s) **491** and/or strain sensor(s) **492** may be positioned on or be integrated with hooks **340**. In an example, the weight sensor(s) **492** may be a scale or form part of a scale. It should be appreciated that the load or weight sensor(s) **490** may be either a load sensor or a weight sensor. Additionally, the load or weight sensor(s) **490** may include a combination of load sensors and weight sensors.

[0117] The level sensor(s) **493** may also be used to monitor the orientation of the cart **210, 310, 410**. For example, level sensor(s) **493** may be used to ensure that the cart is in a horizontal orientation, for example, that each compartment, shelf or tray provides a support surface that resides in a substantially horizontal plane. As noted above, each shelf, compartment or tray may be adjustable and the overall height of carts **210, 310** and **410** may be adjustable. Adjustment may be

via the extension and contraction of the vertical posts of cart **210**, **310**, **410** via adjustments made at wheels **280a** to **280d** or wheels **380a** to **380d**, and/or via adjustment to the height of compartments (e.g., compartments **230**, **240**, **250**) and or shelves. The cart **210**, **310**, **410** may be configured for auto-adjustment or auto-leveling. In other examples, the cart **210**, **310**, **410** may be manually adjusted with adjustment screws located in each of the vertical posts. Depending on the reading from level sensor(s), adjustments may be made to the shelves, the compartments or to the cart in general to position each shelf or compartment in a substantially horizontal orientation when the cart is positioned on an uneven or non-level surface. Properly adjusting the cart, shelves or compartments such that the fluid bags are supported by or on a substantially horizontal surface may improve the accuracy of measurements or readings from level sensor(s) used to determine the mass or volume of fluid in fluid bags or containers.

[0118] FIG. **4** also illustrates that proximity sensor(s) **494** or contact sensor(s) **495** may be employed to determine the presence of fluid bags **120**, drain bags **140**, tubing and/or connectors. In an example, the proximity sensors **494** and/or the contact sensors **495** may be used in combination with light sensor(s) **496** or ozone sensor(s) **497** to ensure that sanitizing devices **468** (e.g., UV lights or ozone generator) are activated when an object to be sanitized (e.g., a fluid bag **120** or connector) is present. Otherwise, the sanitizing devices **468** may be deactivated (or not be activated) if an object to be sanitized is not detected.

Just-in-Time Ordering

[0119] Carts **210**, **310** and **410** may include a communication module **498** that is communicatively connected to a network (e.g., wired or wireless network). Wired communication may be via Ethernet connection, for example. Wireless communication may be performed via any of Bluetooth™, WiFi™, Zigbee®, Z-Wave®, wireless Universal Serial Bus (“USB”), or infrared protocols, or via any other suitable wireless communication technology. Based on information obtained from load or weight sensor(s) **490**, a pressure sensor(s) **491**, a strain sensor(s) **492**, a level sensor(s) **493**, etc., the cart may be configured to order consumables such as fluid bags **120** (e.g., dialysis fluid bags), drain bags **140**, and connectors. The just-in-time ordering functionality of carts **210**, **310** and **410** may continually update delivery schedules based on the sensor information. It should be appreciated that fluid bags, drain bags and connectors are a few examples of consumables that may be ordered over connection established between the cart and the network. These examples are provided for illustrative purposes and are non-limiting. For example, the communication module **498** may be used to order other consumables needed by the patient.

Optimized PD

[0120] Carts **210**, **310** and **410** enable the performance of an “optimized PD” therapy in which different PD dialysis fluids are used to provide an optimized physiological treatment tailored for the patient. An “optimized PD” may include, for example, alternating dwell cycles between a high glucose solution and a no (or low) glucose solution. By performing fluid heating in a compartment or shelf of the cart **210**, **310**, **410**, such that all bags are heated at the same time, cyclers **104** can draw from any of the, e.g., three fluid bags **120a** to **120c** (e.g., dialysis solution bags) during a given fill cycle and can offer the ability to tailor the solution composition per fill cycle.

[0121] Conversely, other systems and “ad hoc” arrangements may require that the first fluid bag **120a** (e.g., dialysis solution bag) is placed on the heating tray **106** of the cycler **104** thus limiting which solution bags the cycler **104** can pull solution from and thus limits the type of APD therapy that can be performed.

[0122] It should be understood that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. It is therefore intended that such changes and modifications are covered by the appended claims.

Claims

- 1.** A mobile dialysis therapy cart comprising: a top shelf; a fluid bag management compartment located beneath the top shelf, the fluid bag management compartment including at least one heating device for heating a fluid bag; at least one temperature sensor located to sense a temperature associated with dialysis fluid located within the fluid bag; and a control unit communicatively coupled to the at least one heating device and the at least one temperature sensor, the control unit programmed to use an output from the at least one temperature sensor to operate the at least one heating device to heat the dialysis fluid within the fluid bag to a desired temperature.
- 2.** The mobile dialysis therapy cart of claim 1, further comprising a drain compartment positioned below the fluid bag management compartment.
- 3.** The mobile dialysis therapy cart of claim 2, further comprising at least one effluent sensor associated with the drain compartment.
- 4.** The mobile dialysis therapy cart of claim 1, further comprising a cyclor compartment positioned below the top shelf, the cyclor compartment sized and shaped to house an automated peritoneal dialysis (“APD”) cyclor.
- 5.** The mobile dialysis therapy cart of claim 4, further comprising at least one sanitizing device, wherein the at least one sanitizing device is a sanitizing light source associated with at least one of the fluid bag management compartment or the cyclor compartment.
- 6.** The mobile dialysis therapy cart of claim 4, further comprising at least one sanitizing device, wherein the at least one sanitizing device is an ozone generator configured to direct ozone containing gas towards at least one of the fluid bag management compartment or the cyclor compartment.
- 7.** The mobile dialysis therapy cart of claim 1, wherein the at least one heating device includes at least one of a resistive plate heater, an inductive heater, and a micro-wave heater.
- 8.** The mobile dialysis therapy cart of claim 1, further comprising at least one of a urea sensor, a creatinine sensor, a conductivity sensor, a turbidity sensor for sensing patient effluent, or a fluid presence sensor for detecting a leak in the fluid bag management compartment.
- 9.** The mobile dialysis therapy cart of claim 1, further comprising at least one of a pressure sensor, a contact sensor, a load or weight sensor, or a strain sensor for sensing a mass or volume of fluid that is delivered to or from a patient.
- 10.** A mobile dialysis therapy cart comprising: a top shelf; a cyclor compartment positioned below the top shelf, the cyclor compartment sized and shaped to house an automated peritoneal dialysis (“APD”) cyclor; a plurality of fluid bag hooks; a drain compartment; and a plurality of wheels.
- 11.** The mobile dialysis therapy cart of claim 10, further comprising at least one sanitizing device configured to sanitize a location of at least one fluid bag held by at least one of the plurality of fluid bag hooks.
- 12.** The mobile dialysis therapy cart of claim 11, wherein the at least one sanitizing device includes a sanitizing light source directed toward the location of the at least one fluid bag.
- 13.** The mobile dialysis therapy cart of claim 10, wherein the top shelf includes a hinge configured to enable a portion of the top shelf to rotate upwards and fold over itself to provide access to the cyclor compartment.
- 14.** The mobile dialysis therapy cart of claim 10, further comprising a drain trolley housed within the drain compartment, wherein the drain compartment includes a door, and wherein the door is configured to fold down away from the drain compartment to provide a ramp for the drain trolley.
- 15.** The mobile dialysis therapy cart of claim 10, further comprising at least one of a urea sensor, a creatinine sensor, and a turbidity sensor associated with the drain compartment.
- 16.** A mobile dialysis therapy cart comprising: a top shelf including a top surface, wherein the top shelf is adjustable and the top surface is hinged; a cyclor compartment positioned below the top shelf, the cyclor compartment being adjustable and being sized and shaped to house an automated peritoneal dialysis (“APD”) cyclor; a fluid bag management compartment including at least two

telescopic sub-shelves, each of the at least two telescopic sub-shelves sized and shaped to support a fluid bag; a drain compartment positioned below the fluid bag management compartment, wherein the drain compartment includes a door that is adapted to open from a top and fold down towards a bottom of the mobile dialysis therapy cart and serve as a ramp surface for a drain trolley that is housed in the drain compartment; and a plurality of wheels.

17. The mobile dialysis therapy cart of claim 16, further comprising a display device, wherein the display device is removable from the mobile dialysis therapy cart while maintaining connectivity with the mobile dialysis therapy cart.

18. The mobile dialysis therapy cart of claim 16, further comprising at least one of a urea sensor, a creatinine sensor, a conductivity sensor and a turbidity sensor associated with the drain compartment.

19. The mobile dialysis therapy cart of claim 16, further comprising a level sensor for sensing an orientation of the mobile dialysis therapy cart, the cyclor compartment, the fluid bag management compartment, or the drain compartment.

20. The mobile dialysis therapy cart of claim 16, further comprising a sanitizing device configured to sanitize a portion of the cyclor compartment, the fluid bag management compartment, or the drain compartment.
